

Md. Farhan Rafiq

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<https://farhan-rafiq.github.io>

Education

- **M.S. (Thesis) in Oceanography** Dec 2025
CGPA: 3.83/4.00 | Shahjalal University of Science and Technology (SUST), Bangladesh
- **B.Sc. in Oceanography** Dec 2023
CGPA: 3.66/4.00 | Shahjalal University of Science and Technology (SUST), Bangladesh

Research Interests

Geographic Information System (GIS), Remote Sensing, Geospatial AI and Predictive Modelling, Natural Hazards and Disaster Resilience, Climate Change and Environmental Monitoring

Research Experience

- **Master's Thesis**
Coastal Urban Flood Risk Modelling in Chattogram: An Integrated Machine Learning and TOPSIS Approach
 - Integrated machine learning algorithms with TOPSIS for urban flood risk modeling.
 - Produced GIS-based flood hazard, vulnerability, and risk maps to identify high-risk zones.
- **Undergraduate Research Project**
Spatial and Temporal Variability of PAR, K_d_{490} , and SST in the Bangladesh EEZ (2016–2022)
 - Analyzed 7 years of MODIS Aqua ocean color and SST data using SeaDAS and ArcGIS.
 - Evaluated spatial and temporal productivity patterns affecting marine ecosystems.
- **Mini Project – Remote Sensing Coursework**
Annual Variability of Chlorophyll-a and Sea Surface Temperature in the Bay of Bengal
 - Applied satellite data to analyze seasonal trends in Chlorophyll-a and SST in the Bay of Bengal.
 - Created detailed geospatial visualizations using SNAP and ArcMap.

Professional Experience

- **Research Assistant** Dec 2025 – Present
Dept. of Oceanography, SUST – Meghna River Estuary Ecohydrological Project
 - Conducting ecohydrological monitoring and GIS-based spatial analysis of estuarine variables.
 - Integrating in situ observations and geospatial analysis for environmental assessment.
- **Research Intern** Mar 2025
Bangladesh Fisheries Research Institute (BFRI), Cox's Bazar, Bangladesh
 - Implemented phytoplankton culture, growth monitoring, and microscopic identification.
 - Performed field sampling, specimen preservation, and isolation planning.
- **Teaching Experience** Oct 2021 – Present
Coaching Instructor & Tutor, Sylhet, Bangladesh
 - Teach Mathematics, Physics, and Science to 35+ O-level and A-level students.
 - Assess student progress, address learning gaps, and improve academic performance.

Conference Presentations

- **International Symposium on Coast, Climate and Community, Sylhet** 17 – 18 Jun 2026
Poster Presentation: *"Coastal Urban Flood Risk Modelling in Chattogram: An Integrated Machine Learning and TOPSIS Approach"*
- **International Conference on Engineering and Natural Sciences, Singapore** 25 – 26 Nov 2025
Oral Presentation (online): *"Spatial and Temporal Dynamics of PAR, Kd₄₉₀, and SST in the Bangladesh EEZ (2016–2022)"*

Technical Skills

- **GIS & Remote Sensing:** ArcGIS Pro, ArcMap, QGIS, SNAP, ODV, Delft3D, ERDAS Imagine
- **Programming Languages:** Python (Pandas, NumPy, Matplotlib), R, Google Earth Engine
- **Oceanographic Instruments:** CTD, ADCP, Water Multi-parameter, AlgaeTorch
- **Data Analysis & Visualization:** Microsoft Excel, Google Sheets, Power BI
- **Research & Writing Tools:** LaTeX, Mendeley, Microsoft Word, Google Docs
- **Survey Tools:** Kobo Toolbox, Microsoft Forms, Google Forms
- **Graphic Design:** Canva Pro, Adobe Illustrator, Microsoft PowerPoint

Honors & Awards

- **National Science and Technology Fellowship** 2024 – 2025
Granted by The Ministry of Science and Technology, Government of Bangladesh

Certifications & Training

- **Going Places with Spatial Analysis**, ESRI
- **Spatial Analysis and Satellite Imagery in GIS**, University of Toronto, Coursera
- **Data Analysis with Python**, IBM, Coursera
- **Cartography**, ESRI

Language Proficiency

- English: Advanced (IELTS Score: 7)
- Bangla: Native

Co-curricular Activities

- **Volunteer** 24 – 25 Mar 2023
International Symposium on Marine Resource Management (ISMRRM), SUST
- Supported symposium logistics, session coordination, and participant facilitation during the event.

References

Dr. Subrata Sarker
Associate Professor and Head,
Dept. of Oceanography, SUST
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